

Introduction

Security Requirements

- Confidentiality: Prevention of unauthorised disclosure of information
- Integrity: Prevention of unauthorised modification of information of processes
- Availability: Prevention of unauthorised withholding of information of resources

Threat Model

- Describe security of a system by the class of attacks it can prevent
- A class of attacks is described by the attacker's goals and capabilities
- Can be used to compare security of systems

Trade-Offs

- Ease of Use: Security mechanisms interfere with working patterns users are originally familiar with
- Performance: Security mechanisms consume more computing resources
- Cost: Security mechanisms are expensive to develop and manage

Vulnerabilities

- CVE: Common Vulnerabilities and Exposures, public list of entries
- Zero Day: Vulnerabilities that are discovered but not published, no reaction time
- Adversarial thinking: Thinking like attacker when designing secure systems

Notions

- Threat: Set of circumstances that has the potential to cause loss or harm
- Vulnerability: A weakness in the system
- Control: A means to counter threats
- A threat is blocked by control of a vulnerability

Encryption

Definitions

- Symmetric Key Encryption Scheme: Consists of encryption and decryption algorithm
- Correctness: $D_k(E_k(x)) = x$
- Security (Informally): Eavesdropper cannot derive useful information about key or plaintext even if they can probe
- Cryptography: Study of techniques in securing communication in the presence of attackers who have access to it

Thread Model

- Attacker's Goals:
 - Total Break: Attacker wants to find the key
 - Partial Break: Many definitions, for e.g. attacker wants to decrypt a specific ciphertext
 - Distinguishability: Attacker can distinguish ciphertexts between one plaintext and another
 - An attacker who can achieve total break can also achieve partial break and distinguishability
- Attacker's capability:
 - Ciphertext only: Attacker has a collection of ciphertexts, and may know some properties of it
 - Known plaintext: Attacker has a collection of plaintexts and corresponding ciphertexts
 - Chosen plaintext attack (CPA): Attacker has access to encryption oracle, which they can feed any plaintext to to get the corresponding ciphertext

- Chosen ciphertext attack (CCA2): Same as CPA but attacker can choose ciphertext to use with decryption oracle to get plaintext
- If a cipher can defend against a decryption oracle, then it can defend against all other weaker forms
- If a cipher is secure against CCA2, it is also secure against CPA

Classical Ciphers

Substitution Cipher

- The plaintext and ciphertext are strings over a set of symbols
- The key is a substitution table which is a bijective function from the set of symbols to itself
- The key space is the set of all possible keys with the size being the total number of possible keys
- The key size is the number of bits required to represent a key
- Encryption: Given plaintext $X = x_1x_2...x_n$ and key S , output ciphertext $E_S(X) = S(x_1)S(x_2)...S(x_n)$
- Decryption: Given ciphertext $C = c_1c_2...c_n$ and key S , output plaintext $D_S(C) = S^{-1}(c_1)S^{-1}(c_2)...S^{-1}(c_n)$
- Simple attack is to perform an exhaustive search on the key space
- To be secure, exhaustive search must be computationally infeasible
- Running time depends on the size of the key space, needing $n!$ loops where n is the size of the alphabet
- Known plaintext attack: Attacker has a plaintext and ciphertext, and can use it to reverse engineer some entries in the key
- Example would be headers in packets where starting is fixed
- Ciphertext only attack: Attacker knows plaintext is an English sentence, so they brute force all keys on the ciphertext, only considering when the result has English words
- Frequency analysis: Attacker can guess plaintext by mapping letters to their most frequently occurring counterparts
- Substitution cipher is not secure under both types of attacks

Permutation Cipher

- Group plaintext into blocks of t characters then apply a permutation function to each block, where the permutation is a bijective function from $1...t$ to itself
- $p = (p_1, p_2, ..., p_t)$ which shifts the character i to position p_i
- Fails miserably under known-plaintext attack
- Performing substitution or substitution twice does not increase difficulty of attack, but interleaving them can make it more difficult

One Time Pad

- Encryption: Given n bit plaintext $x_1, ..., x_n$ and n bit key $k_1, ..., k_n$, $C = (x_1 \oplus k_1), ..., (x_n \oplus k_n)$
- Decryption: Given n bit ciphertext $c_1, ..., c_n$ and n bit key $k_1, ..., k_n$, $X = (c_1 \oplus k_1), ..., (c_n \oplus k_n)$
- Key cannot be reused, but must be same size as plaintext
- Correctness is proven by $(x \oplus k) \oplus k = x \oplus (k \oplus k) = (x \oplus 0) = x$
- Attacker can derive key with a plaintext and ciphertext pair, but it is useless since keys are only used once
- Exhaustive search does not work, and one time pads do not leak any information of plaintext

Modern Ciphers

DES / Exhaustive Search

- Quantify security of encryption scheme by length of key, a scheme with larger keys is more secure wrt exhaustive search
- DES uses 56 bit keys, but exhaustive search is now feasible due to improvements in computing power

AES (Advanced Encryption Standard)

- Block length of 128 while key length can be 128, 192, or 256 bits
- Currently no known attacks

Block Cipher & Mode-of-Operations

- DES and AES are block ciphers which are designed for fixed size inputs and outputs
- Block cipher is applied to each block in a large plaintext
- Mode-of-operation specifies how encryption is extended from a single block to multiple blocks
- ECB (Electronic Code Book) mode: Plaintext is divided into blocks before block cipher is applied to each block all with the same key
- ECB leaks information, and is deterministic, so additional mechanisms are required to prevent leakage
- CBC (Cipher Block Chaining): Initialisation Vector arbitrarily chosen
- Encryption: $c_0 = IV, c_i = E_k(x_i \oplus c_{i-1}), i > 0$ is used to generate ciphertext y
- Decryption: $P_1 = D_k(C_1) \oplus IV, P_i = D_k(C_i) \oplus C_{i-1}, i > 0$
- CTR (Counter Mode): Each block x_i uses $IV + i$, at stream cipher
- Encryption: $c_i = x_i \oplus E_k(IV + i)$
- Decryption: $p_i = c_i \oplus E_k(IV + i)$
- GCM (Galois Counter Mode): Authenticated Encryption, produces an extra tag for authentication, secure in presence of decryption oracle

Stream Cipher and IVs

- Stream cipher has an Initialisation Vector (IV)
- Pseudorandom sequence is generated from secret key and IV
- Final ciphertext contains IV followed by output of OTP encryption
- For decryption, IV is extracted from ciphertext, so that same pseudorandom sequence can be generated and used to obtain plaintext
- IV should be changed to prevent attacker from eavesdropping and decoding
- If IVs are different in two different instances of encryption, the two pseudorandom sequences will also be different, and xoring them will not cancel them out
- CBC mode also needs IV for non-determinism

Attacks

Triple DES & MITM

- DES is not secure wrt computing power, can be improved using multiple encryptions
- Meet in the Middle: Attacker brute forces encryption from plaintext with first key and decryption from ciphertext with second key
- Tries to find match in the middle
- Instead of 2^{2k} , is now 2^{k+1} time
- Tradeoff between time and space: Can exhaustively select last s bits of each so that less storage space at 2^{k-s+1} but number of operations increases to 2^{s+k+1}
- 3DES: Use triple encryption with 2 keys to make it harder to break, e.g. $E_{k1}(E_{k2}(E_{k1}(x)))$ or $E_{k1}(D_{k2}(E_{k1}(x)))$

Padding Oracle

- An oracle is a query answer system
- If length of plaintext is not a multiple of block size, remaining bytes need to be padded, but receiver also needs to know how much padding was used
- PKCS#7 is a padding standard where padding of x bytes is done with x copies of x , if block is full an extra block of the block size is added
- The padding oracle knows the key and outputs yes if the ciphertext is in the correct padding format
- AES CBC is not secure against padding oracle attack

- Attacker knows IV, C, and padding format; padded with **3** bytes in this case
- Attacker calculates iv' from iv by xoring the last 4 bytes where t is set to some value i.e. **0000t777** to force the padding to flip to **04**
- Attacker feeds padding oracle with (iv', c)
- If oracle outputs YES, then $x_5 \oplus t$ must be 04, and x_5 is **04 $\oplus$ t**
- If not, try with another t value
- Can be extended to find all plaintexts, only information needed is initial padding length which is easy to determine
- Find by flipping from left to right and checking padding
- Solution is to deny access to such an oracle, or avoid padding in the first place if possible
- CTR mode is also vulnerable to padding oracle attack

Cryptography Pitfalls

- Wrong choice of IV: e.g. Deriving IV from filename, which can be repeated across different files, should not be same for different ciphertexts
- Reusing one time pads
- Predictable Secret Generation: Using a fixed seed
- Designing your own cipher or modifying an existing scheme; use well-accepted algorithms and libraries
- Reliance on Obscurity: Kerckhoff's principle says a system should be secure even if everything about the system other than secret key is known
- Security through obscurity is achieving security by hiding system design, but once exposed becomes useless

Authentication (Password)

- Authentication: Assuring that communicating entity or origin of information is the one it claims to be, implies integrity
- Entity Authentication: For connection-oriented communication, verifying authenticity of entities involved
- Data-Origin Authentication: For connectionless communication, verifying origin of a piece of information
- Credential: Information bound to owner which is required for authentication, is authentic with owner can convince verifier it has access to it

Passwords

Password System

- Bootstrapping: User and server establish common password, and server keeps a password file recording the users identity and corresponding password
- Authentication: Server authenticates an entity by ensuring it gives the correct password
- Replay attack: Sniffed information can be replayed to impersonate a user
- Password Reset: User can reset password if necessary, but authentication has to be done first
- Methods include recovery emails and security questions

Password Attacks

- Attacker can intercept password during bootstrapping or exploit usage of default passwords
- Dictionary attack tests all passwords stored in a dictionary and tries all combinations as a form of exhaustive search
- Online dictionary attack: Attacker must interact with authentication system while searching (29 bits minimum, 49 bits recommended)
- Offline dictionary attack: Attacker obtains some information about password, then carries out search using it without interacting with system through methods such as testing hashes (128 bits minimum)
- Social engineering can be carried out to construct a dictionary
- Passwords can be stolen through sniffing in various forms such as key loggers or viruses

- Phishing attacks are social engineering attacks where the goal is to trick the user to visit a spoofed website and key in their login information
- Spear phishing is target to a particular small group of users
- Cache usage on platforms like a browser can also leak passwords
- If the system administrators account is compromised, password files could also be stolen
- Prevention can be done through education and blacklisting phishing sites and using strong passwords
- Password Entropy: Measure of strength of password as a measurement of randomness
 - $-\sum_{i=1}^N p_i \log_2 p_i$, maximised when probability is uniform
- Systems can also have login timeouts or check password strength or require regular password changes, and should also have protected password files using hashing and salting
- Passwords generated by human, secret keys generated by machines; key derivation function (KDF) uses password as source for crypto secret key e.g. SHA3

Biometrics

- Uses physical characteristics of a person for authentication
- During enrollment, a template of a user's biometric data is stored, for verification using a matching algorithm
- Biometric data acts as password, but has a probability of error
- Threshold adjusted for matching algorithm
- Can be spoofed, some systems include liveness detection through methods such as temperature sensors

n -factor Authentication

- Uses more than one authentication factor
- An example is a password and SMS verification or OTP token
- OTP token shares secret with server
- Time Based: Based on shared secret and current time interval, a password is generated
- Sequence Based: An event triggers change of password

Authenticity (Data Origin: MAC & Signature)

Crypto Primitive: PKC (Public Key Crypto)

- Asymmetric key scheme uses different keys for encryption and decryption
- Encryption is done with public key while decryption is done with private key
- Only holder of private key can decipher ciphertext
- Must be difficult to get private key from public key
- Only requires a broadcast channel, does not need a secure channel for sharing of symmetric keys, also uses less keys in total
- Authentication is an extremely important part of PKC
- Many PKC schemes represent data as integers to which arithmetic operations are applied

RSA

- User chooses large prime p, q and computes $n = pq, \varphi(n) = (p - 1)(q - 1)$
- User randomly chooses an encryption exponent e with $\gcd(e, (p - 1)(q - 1)) = 1$
- User finds decryption exponent d where $de \bmod (p - 1)(q - 1) = 1$
- (n, e) is the public key while (n, d) is the private key
- $c = m^e \bmod n$ while $m = c^d \bmod n$
- $D(E(m)) = m, m^{ed} \bmod n = m$ since $m^{de} \bmod n = m \bmod n = m$
- e and d can have their roles swapped so everyone can decrypt but only owner can encrypt
- Common public key is 65537 which is small enough
- Difficulty comes from factorising the large value n

- Post-Quantum Cryptography: PKC that are secure against quantum computers, unlike RSA which will be broken
- IVs needed to get different ciphertexts from the same plaintext at different times
- Not known whether getting plaintext from ciphertext and public key is as difficult as factorisation
- Textbook RSA has to be modified as attacks can use the homomorphic property to extract one bit of information
- 128 bit AES and 3072 bit RSA have equivalent key strength, but RSA is significantly slower
- PKC is useful for authentication
- Typically encrypt AES key with RSA, while using AES to encrypt file instead, send concatenated ciphertext
- Other schemes: Discrete log-based PKC such as ElGamal or Paillier Encryption or Lattice-based PKCs

Data Authenticity (Digest), Unkeyed

- A hash is a function that takes an arbitrary large message as input and outputs a fixed size digest
- Collision-Resistant: Difficult to find different m_1, m_2 such that $h(m_1) = h(m_2)$
- 2nd Preimage: Given m_1 , find m_2 such that $h(m_1) = h(m_2)$
- Preimage: Given y , find m such that $h(m) = y$ (one-way)
- Popular hash functions are the SHA or MD family
- SHA1 and MD5 collisions have been found, use SHA256 or SHA3 instead
- Application: Verifying authenticity of file download, attacker tries to find second preimage that is malicious

Data Origin Authenticity (MAC), Keyed

- A keyed-hash takes an arbitrarily large message and secret key as input and outputs a fixed size message authentication code
- Even after seeing multiple valid pairs of messages and MAC, it is still difficult for an attacker to forge an MAC
- CBC-MAC: Start with IV **0** and repeatedly xor with x_i followed by throwing it in e_k
- Application: When there is no secure channel to deliver digest
- Typically MAC is appended to a file and stored together as authentication tag

Data Origin Authenticity (Signature), Asymmetric Key

- The public key version of MAC is a signature
- Owner uses private key to generate signature which public can verify using public key
- Easier to manage than MAC
- Non-repudiation: Assurance that someone cannot deny previous actions
- Popular schemes use RSA for signing and verifying
- Signature schemes consist of unkeyed hash followed by signing and verifying algorithm
- With RSA, $s = RSA_E(key^-, H(x))$, (X, s) can be verified by checking $H(X) = RSA_D(key^+, s)$
- Some signature schemes do not use encryption in their scheme

Attacks and Pitfalls

- Birthday Attacks: Repeatedly generate messages until probability of finding collision is high
- Suppose we have $M = 1.17 \cdot 2^{64}$ messages and each has a value from $T = 2^{128}$, the probability of collision is $1 - e^{(-\frac{M^2}{2T})}$, this gives $P > 0.5$, basically half the bits
- For a set of k distinct elments of n bits, select m of them and put them in another set. The probability of non-empty intersection is $> 1 - 2.7^{-km2^{-n}}$
- Using encryption for authentication: Encryption provides confidentiality, but not integrity or authenticity